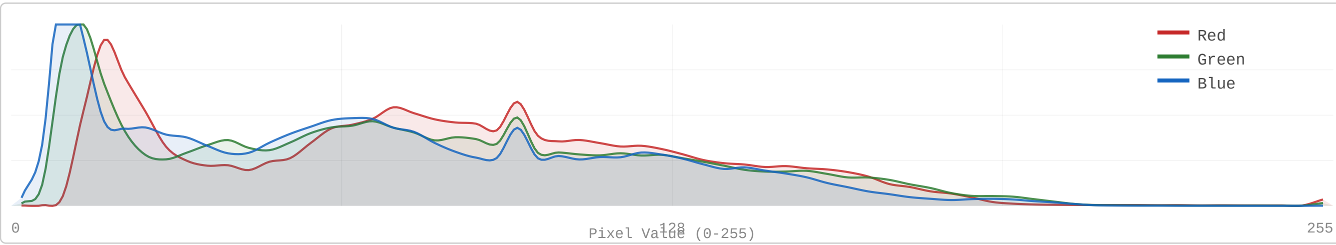
Decorrelation Gap: KODIM17

512x768 | 24-bit RGB | PNG (lossless)

Sailboat in Harbor at Sunset | Near-degenerate (PC1 = 97.78%)

1. RGB Channel Distribution



2. Inter-Channel Correlation: RGB vs YCbCr

RGB (Original)

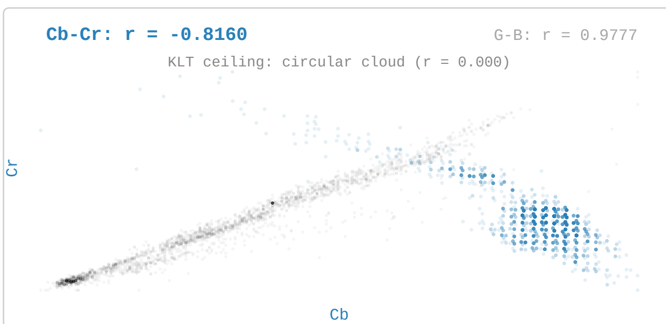
	R	G	B
R	1.0000	0.9852	0.9334
G	0.9852	1.0000	0.9777
B	0.9334	0.9777	1.0000

Cb-Cr:
-0.816
(KLT ceiling = 0.000)

YCbCr (BT.601)

	Y	Cb	Cr
Y	1.0000	-0.1756	-0.3062
Cb	-0.1756	1.0000	-0.8160
Cr	-0.3062	-0.8160	1.0000

3. Cb vs Cr Scatter with G-B background overlay



4. Pairwise Decorrelation Gap

Channel Pair	RGB	YCbCr (BT.601)	Residual |r|
R-G → Y-Cb	0.985194	-0.175638	0.1756
R-B → Y-Cr	0.933405	-0.306169	0.3062
G-B → Cb-Cr	0.977749	-0.815979	0.8160
Avg |r|	0.965449	0.432596	0.4326

Decorrelation Gap: KODIM17

512x768 | 24-bit RGB | PNG (lossless)

Sailboat in Harbor at Sunset | Near-degenerate (PC1 = 97.78%)

5. Decorrelation Efficiency & Blue Independence

Decorrelation Efficiency

55.2% Moderate

0% BT.601 leaves all — 100% KLT ceiling

Blue Channel Independence (Baetzel 2026a)

3.6%

0% Fully predictable — 100% Fully independent

6. 4:2:0 Chroma Subsampling — Information Cost

Channel	PSNR (dB)
R	47.07 dB
G	50.52 dB
B	43.48 dB
Overall	46.11 dB

Method: BT.601 RGB→YCbCr | 2x box downsample Cb,Cr | nearest upsample | BT.601 inverse

7. PCA Baseline Reference (Baetzel 2026a)

Eigendecomposition

Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5578	-0.6048	-0.5684	6856.26
PC2	0.6771	0.0645	-0.7330	150.75
PC3	-0.4800	0.7938	-0.3735	4.94

8. PCA Derived Metrics

Condition Number

1388.78

PC1/PC2: 45.48 | PC2/PC3: 30.54

Blue Channel Independence

3.6%

Variance not captured by PC1

PC1 Dominant Channel

G

Loading: G (0.60)

9. Redundancy Profile

Classification: Near-degenerate | CN: 1388.78 | PC1: 97.8%

RGB Correlation: Avg |r| = 0.965449 | R-G: 0.9852 | R-B: 0.9334 | G-B: 0.9777

YCbCr Residual: Avg |r| = 0.432596 | Y-Cb: -0.1756 | Y-Cr: -0.3062 | Cb-Cr: -0.8160

Efficiency: 55.2% of KLT ceiling | Gap: 0.4326 avg |r| remaining

Blue Channel: 3.6% independent | 4:2:0 PSNR: 46.11 dB

References

[1] Baetzel, J. (2026a). "Statistical Characterization of Inter-Channel Redundancy Structure"
[2] Wallace, G.K. "The JPEG still picture compression standard." IEEE Trans. CE 38(1), 1992.
[3] ITU-R Recommendation BT.601: Studio Encoding Parameters, 1982.
[4] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.